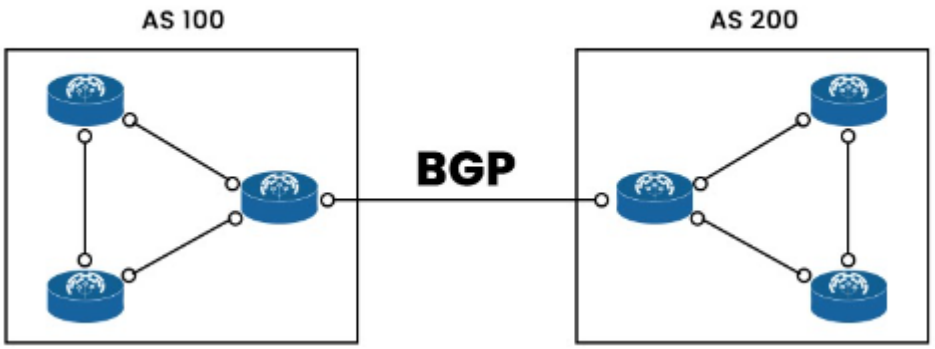
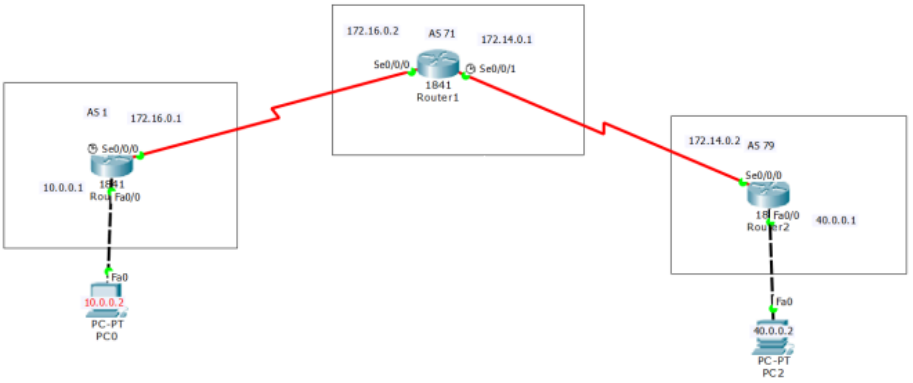
Introduction to BGP Protocol

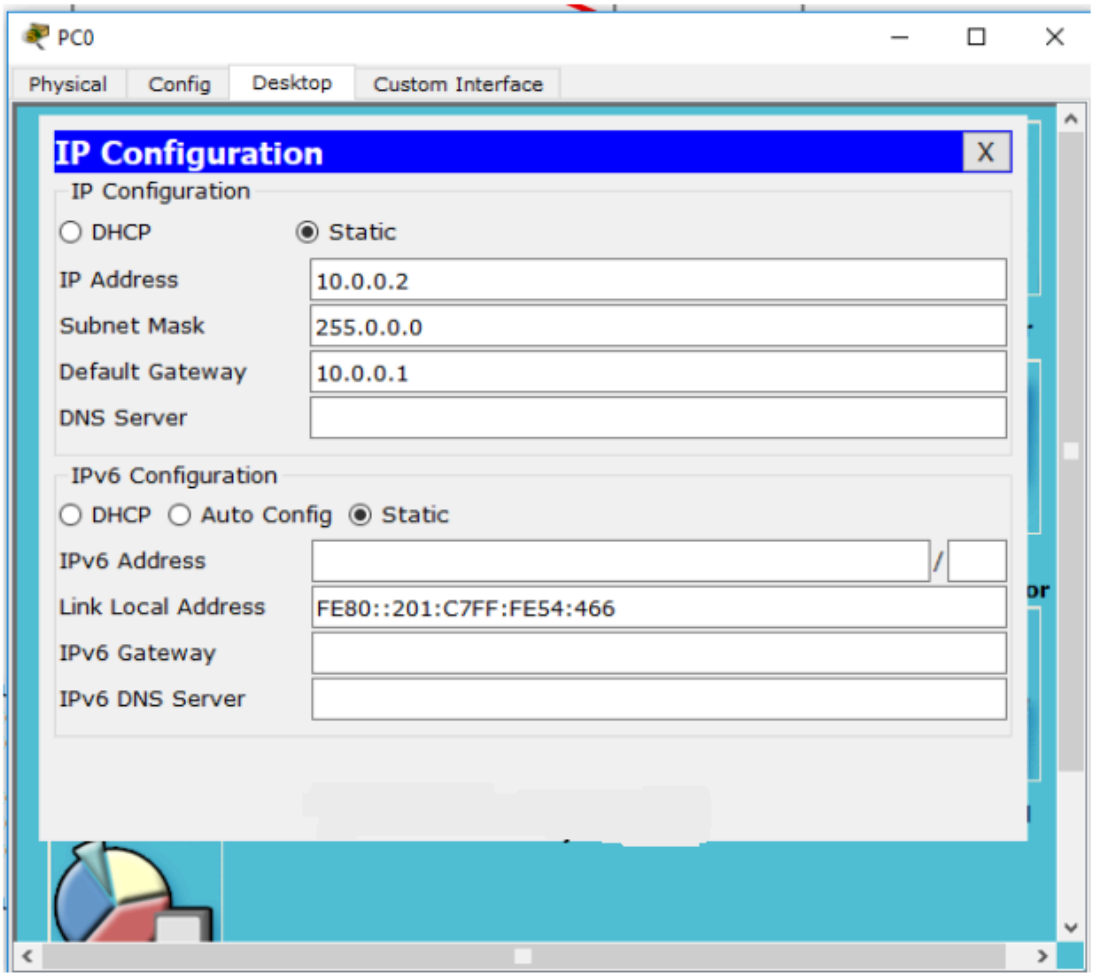
Border Gateway Protocol (BGP) is a critical protocol used in the internet routing infrastructure to facilitate communication between different autonomous systems (AS). Unlike interior gateway protocols (IGP) such as OSPF or EIGRP, which focus on routing within a single autonomous system, BGP is an exterior gateway protocol designed for routing between autonomous systems. It plays a crucial role in ensuring efficient and reliable routing across the global internet.

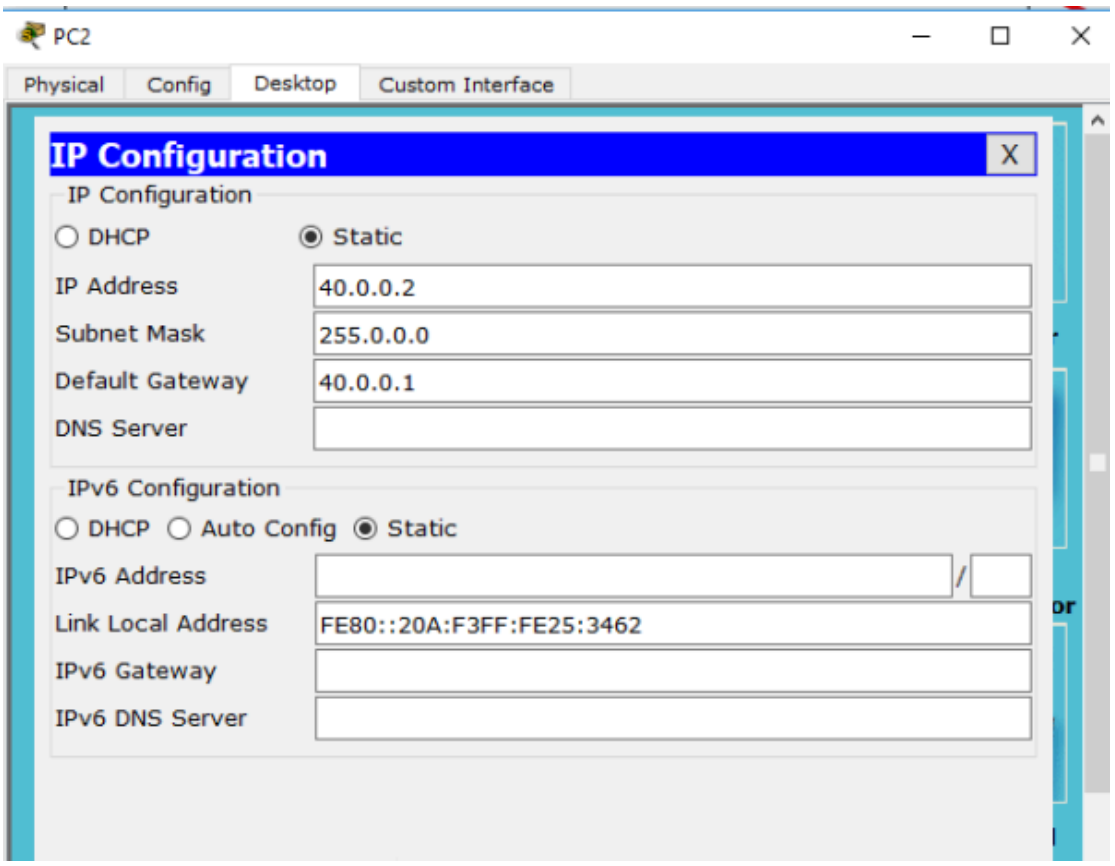


Key Features of BGP:
Path Vector Protocol: BGP is a path vector protocol, which means it makes routing decisions based on the path attributes (such as AS path, next hop, etc.) rather than just the shortest path to a destination. This enables BGP to make complex routing decisions and support policy-based routing.
Policy-Based Routing: BGP allows network administrators to implement sophisticated routing policies. These policies can control how routes are learned, advertised, and manipulated based on various criteria like AS paths, prefix lengths, communities, and local preferences.
Reliability and Scalability: BGP is designed to handle the scale and complexity of the global internet. It can manage a large number of routes and adapt to changes in network topology dynamically while maintaining stable routing tables.

Lab Task:
In this lab we will configure bgp protocol using Cisco packet tracer.We will implement the following topology:




PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
10.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.1
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::201:C7FF:FE54:466
IPv6 Gateway
IPv6 DNS Server



Step 3: bgp configuration on Router R1:

R1(config)#router bgp 1
R1(config-router)#neighbor 172.16.0.2 remote-as 71
R1(config-router)#network 10.0.0.0 mask 255.0.0.0
R1(config-router)#exit
R1(config)#do write
Building configuration...[OK]
R1(config)#

Step 4: bgp configuration on Router R2:

R2(config)#router bgp 71
R2(config-router)#neighbor 172.16.0.1 remote-as 1
R2(config-router)#neighbor 172.14.0.2 remote-as 79
R2(config-router)#network 40.0.0.0 mask 255.0.0.0
R2(config-router)#exit
R2(config)#do write
Building configuration...[OK]
R2(config)#

Step 5: bgp configuration on Router R3:

R3(config)#router bgp 79
R3(config-router)#neighbor 172.14.0.1 remote-as 71
R3(config-router)#network 40.0.0.0 mask 255.0.0.0
R3(config-router)#exit
R3(config)#do write
Building configuration...[OK
R3(config)#

Step 6: bgp configuration Testing and troubleshooting.

For bgp testing we will ping both pc and check the network communication.

Now I am on PC2:
PC>ipconfig
PC>ping 10.0.0.2
Step 7: check bgp route on router R1:
R1#show ip route
Step 8: Check whether bgp protocols configure or not on Routre R1:
R1#show ip protocols
Step 9: Show BGP Status
R1#show ip bgp summary
Show bgp neighbors status:
R1#show ip bgp neighbors
Similarly we check bgp route on Router R2:
R2#show ip route
R2#show ip protocols
Similarly check bgp route on Router R3:
R3#show ip route
R3#show ip protocols